

Name : Mahathi A J

RollNo: 241801148

Operating System

1.1 GENERAL PURPOSE COMMANDS

1. The 'date' command
2. The 'echo' command
3. The 'cal' command:
4. The 'bc' command:
5. The 'who' command
6. The 'who am i' command
7. The 'id' command
8. The 'tty' command
9. The 'clear' command
10. The 'man' command
11. The 'ps' command
12. The 'uname' command

```
[mahathi@localhost ~]$ date
Saturday 17 January 2026 05:35:16 PM IST
[mahathi@localhost ~]$ date +%m
01
[mahathi@localhost ~]$ date +%h
Jan
[mahathi@localhost ~]$ date +%d
17
[mahathi@localhost ~]$ date +%y
26
[mahathi@localhost ~]$ date +%H
17
[mahathi@localhost ~]$ date +%M
36
[mahathi@localhost ~]$ date +%S
51
[mahathi@localhost ~]$ echo
[mahathi@localhost ~]$ echo "Hare Krishna"
Hare Krishna
[mahathi@localhost ~]$ cal Feb 2026
    February 2026
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28

[mahathi@localhost ~]$ bc -l
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
17/8
2.12500000000000000000
6/9
.66666666666666666666
quit
[mahathi@localhost ~]$ who
mahathi  seat0          2026-01-17 14:38 (login screen)
mahathi  tty2            2026-01-17 14:38 (tty2)
[mahathi@localhost ~]$ who am i
[mahathi@localhost ~]$ id
uid=1000(mahathi) gid=1000(mahathi) groups=1000(mahathi) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[mahathi@localhost ~]$ tty
/dev/pts/0
```

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Operating System

```
[mahathi@localhost ~]$ man
What manual page do you want?
For example, try 'man man'.
[mahathi@localhost ~]$ man man
[mahathi@localhost ~]$ ps
  PID TTY          TIME CMD
  3109 pts/0    00:00:00 bash
  3424 pts/0    00:00:00 ps
[mahathi@localhost ~]$ ps -e
  PID TTY          TIME CMD
    1 ?           00:00:01 systemd
    2 ?           00:00:00 kthreadd
    3 ?           00:00:00 pool_workqueue_
    4 ?           00:00:00 kworker/R-rcu_g
    5 ?           00:00:00 kworker/R-sync_
    6 ?           00:00:00 kworker/R-slub_
    7 ?           00:00:00 kworker/R-netns
   11 ?           00:00:00 kworker/R-mm_pe
   13 ?           00:00:00 rcu_tasks_kthre
   14 ?           00:00:00 rcu_tasks_rude_
   15 ?           00:00:00 rcu_tasks_trace
   16 ?           00:00:00 ksoftirqd/0
   17 ?           00:00:00 rcu_preempt
   18 ?           00:00:00 rcu_exp_par_gp_
   19 ?           00:00:00 rcu_exp_gp_kthr
   20 ?           00:00:00 migration/0
   21 ?           00:00:00 idle_inject/0
   23 ?           00:00:00 cpuhp/0
   25 ?           00:00:00 kdevtmpfs
   26 ?           00:00:00 kworker/R-inet_
   27 ?           00:00:00 kauditd
   28 ?           00:00:00 khungtaskd
   29 ?           00:00:00 oom_reaper
   30 ?           00:00:00 kworker/R-write
   31 ?           00:00:01 kcompactd0
   32 ?           00:00:00 ksmd
   33 ?           00:00:01 khugepaged
   34 ?           00:00:00 kworker/R-crypt
   35 ?           00:00:00 kworker/R-kbloc
   36 ?           00:00:00 kworker/R-kinte
```

```
mahathi 2186 0.0 0.3 669668 6588 ? Ssl 14:38 0:00 /usr/libexec/gsd-rfkill
mahathi 2187 0.0 0.3 448060 6312 ? Ssl 14:38 0:00 /usr/libexec/gsd-screensaver-proxy
mahathi 2188 0.0 0.6 678984 10684 ? Ssl 14:38 0:00 /usr/libexec/gsd-sharing
mahathi 2193 0.0 0.7 603804 12392 ? Ssl 14:38 0:01 /usr/libexec/gsd-smartcard
mahathi 2196 0.0 0.5 530568 9192 ? Ssl 14:38 0:00 /usr/libexec/gsd-sound
mahathi 2202 0.0 0.4 232052 7352 ? Sl 14:38 0:00 /usr/libexec/gsd-disk-utility-notify
mahathi 2204 0.0 0.4 671476 7444 ? Ssl 14:38 0:00 /usr/libexec/gsd-usb-protection
mahathi 2213 0.0 1.3 590812 24440 ? Ssl 14:38 0:00 /usr/libexec/gsd-wacom
mahathi 2234 0.0 2.3 843872 40716 ? Sl 14:38 0:00 /usr/libexec/evolution-data-server/evolution-alarm-notify
mahathi 2264 0.0 1.4 2796544 25868 ? Ssl 14:38 0:00 /usr/bin/gjs /usr/share/gnome-shell/org.gnome.ScreenSaver
mahathi 2276 0.0 1.1 600788 19344 ? Sl 14:38 0:00 /usr/libexec/gsd-printer
mahathi 2284 0.0 0.4 375268 7308 ? Sl 14:38 0:00 /usr/libexec/ibus-engine-simple
mahathi 2400 0.0 0.9 762724 16056 ? Ssl 14:38 0:00 /usr/libexec/xdg-desktop-portal
mahathi 2407 0.0 0.3 670072 6740 ? Ssl 14:38 0:00 /usr/libexec/xdg-document-portal
root 2411 0.0 0.1 2664 1892 ? Ss 14:38 0:00 fusermount -o rw,nosuid,nodev,fsname=portal,auto_unmount,subtype=portal -- /run/user/1000/d
mahathi 2417 0.0 1.5 897428 27620 ? Ssl 14:38 0:00 /usr/libexec/xdg-desktop-portal-gnome
mahathi 2467 0.0 1.4 591516 25760 ? Ssl 14:38 0:00 /usr/libexec/xdg-desktop-portal-gtk
root 2479 0.0 1.4 640916 25328 ? Ssl 14:39 0:01 /usr/libexec/fwupd/fwupd
mahathi 2625 0.0 0.3 374740 6364 ? Ssl 14:39 0:00 /usr/libexec/gvfsd-metadata
root 2957 0.0 0.0 0 0 ? I< 15:12 0:00 [kworker/R-tls-s]
mahathi 3091 0.0 2.9 774064 52024 ? Ssl 15:49 0:02 /usr/libexec/gnome-terminal-server
mahathi 3109 0.0 0.3 224124 5704 pts/0 Ss 15:49 0:00 bash
root 3160 0.0 0.0 0 0 ? I 16:00 0:00 [kworker/u4:2-events_unbound]
root 3189 0.0 0.0 0 0 ? I 16:38 0:00 [kworker/u4:1-events_unbound]
root 3233 0.1 0.0 0 0 ? I 17:19 0:01 [kworker/0:1-events]
root 3234 0.0 0.0 0 0 ? I 17:24 0:00 [kworker/0:2-ata_sff]
root 3251 0.0 0.0 0 0 ? I< 17:29 0:00 [kworker/0:0H-kblockd]
root 3254 0.1 0.0 0 0 ? I 17:30 0:00 [kworker/u4:0-events_unbound]
root 3255 0.0 0.0 0 0 ? I 17:30 0:00 [kworker/0:3-ata_sff]
root 3326 0.0 0.0 0 0 ? I 17:36 0:00 [kworker/0:0-ata_sff]
mahathi 3434 0.0 0.3 235404 6960 pts/0 R+ 17:40 0:00 ps -aux
[mahathi@localhost ~]$ uname -a
Linux localhost.localdomain 5.14.0-654.el9.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Dec 19 08:34:59 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
[mahathi@localhost ~]$ uname -m
x86_64
[mahathi@localhost ~]$ uname -n
localhost.localdomain
[mahathi@localhost ~]$ uname -r
5.14.0-654.el9.x86_64
[mahathi@localhost ~]$ uname -s
Linux
[mahathi@localhost ~]$ uname -v
#1 SMP PREEMPT_DYNAMIC Fri Dec 19 08:34:59 UTC 2025
[mahathi@localhost ~]$ uname -a
Linux localhost.localdomain 5.14.0-654.el9.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Dec 19 08:34:59 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
[mahathi@localhost ~]$ █
```

Name : Mahathi A J

RollNo: 241801148

Operating System

1.2 DIRECTORY COMMANDS

1. The 'pwd' command:
2. The 'mkdir' command:
3. The 'rmdir' command:
4. The 'cd' command:
5. The 'ls' command:

```
[mahathi@localhost ~]$ pwd
/home/mahathi
[mahathi@localhost ~]$ mkdir mahathi148
[mahathi@localhost ~]$ ls
Desktop  Documents  Downloads  krishna  mahathi148  Music  Pictures  Public  Templates  Videos
[mahathi@localhost ~]$ cd mahathi148
[mahathi@localhost mahathi148]$ ls
[mahathi@localhost mahathi148]$ ls -l
total 0
[mahathi@localhost mahathi148]$ ls -a
.  ..
[mahathi@localhost mahathi148]$
```

```
[mahathi@localhost mahathi148]$ cat>maha
hi, this is mahathi. also known as leo das.
[mahathi@localhost mahathi148]$ cat maha
hi, this is mahathi. also known as leo das.
[mahathi@localhost mahathi148]$ cap maha aids
bash: cap: command not found...
[mahathi@localhost mahathi148]$ cp maha aids
[mahathi@localhost mahathi148]$ rm -f maha
[mahathi@localhost mahathi148]$ ls
aids
[mahathi@localhost mahathi148]$ mv maha aids
mv: cannot stat 'maha': No such file or directory
[mahathi@localhost mahathi148]$ cat>maha
minecraft
[mahathi@localhost mahathi148]$ mv maha aids
[mahathi@localhost mahathi148]$ file aids
aids: ASCII text
[mahathi@localhost mahathi148]$ file maha
maha: cannot open 'maha' (No such file or directory)
[mahathi@localhost mahathi148]$ wc cs
wc: cs: No such file or directory
[mahathi@localhost mahathi148]$ wc aids
 1  1 10 aids
[mahathi@localhost mahathi148]$ ls>aids
[mahathi@localhost mahathi148]$ who|wc -l
2
[mahathi@localhost mahathi148]$ who|tee sample|wc -l
2
[mahathi@localhost mahathi148]$ ls r**
ls: cannot access 'r**': No such file or directory
[mahathi@localhost mahathi148]$ ls?kkk
bash: ls?kkk: command not found...
[mahathi@localhost mahathi148]$ ls ?kkk
ls: cannot access '?kkk': No such file or directory
[mahathi@localhost mahathi148]$ ls[a-m]
bash: ls[a-m]: command not found...
[mahathi@localhost mahathi148]$ ls[!a-m]
bash: ls[!a-m]: command not found...
[mahathi@localhost mahathi148]$ ls [a-v]
ls: cannot access '[a-v]': No such file or directory
[mahathi@localhost mahathi148]$ ls aids
aids
[mahathi@localhost mahathi148]$ ls aids -rwxr-xr--l Lak std 1525 jan10 12:10 aids
```

Name : Mahathi A J

RollNo: 241801148

Operating System

1.3 FILE HANDLING COMMANDS

1. The 'cat' command:
2. The 'Display contents of a file' command:
3. The 'cp' command:
4. The 'rm' command:
5. The 'mv' command:
6. The 'file' command:
7. The 'wc' command:
8. The 'Directing output to a file' command:
9. The 'pipes' command:
10. The 'tee' command:
11. The 'Metacharacters of unix' command:
12. The 'File permissions' command:
13. The 'chmod' command:
14. The 'Octal Notations' command:

```
hi, this is mahathi. also known as leo das.
[mahathi@localhost mahathi148]$ cap maha aids
bash: cap: command not found...
[mahathi@localhost mahathi148]$ cp maha aids
[mahathi@localhost mahathi148]$ rm -f maha
[mahathi@localhost mahathi148]$ ls
aids
[mahathi@localhost mahathi148]$ mv maha aids
mv: cannot stat 'maha': No such file or directory
[mahathi@localhost mahathi148]$ cat>maha
minecraft
[mahathi@localhost mahathi148]$ mv maha aids
[mahathi@localhost mahathi148]$ file aids
aids: ASCII text
[mahathi@localhost mahathi148]$ file maha
maha: cannot open 'maha' (No such file or directory)
[mahathi@localhost mahathi148]$ wc cs
wc: cs: No such file or directory
[mahathi@localhost mahathi148]$ wc aids
 1  1 10 aids
[mahathi@localhost mahathi148]$ ls>aids
[mahathi@localhost mahathi148]$ who|wc -l
 2
[mahathi@localhost mahathi148]$ who|tee sample|wc -l
 2
[mahathi@localhost mahathi148]$ ls r**
ls: cannot access 'r**': No such file or directory
[mahathi@localhost mahathi148]$ ls?kkk
bash: ls?kkk: command not found...
[mahathi@localhost mahathi148]$ ls ?kkk
ls: cannot access '?kkk': No such file or directory
[mahathi@localhost mahathi148]$ ls[a-m]
bash: ls[a-m]: command not found...
[mahathi@localhost mahathi148]$ ls[!a-m]
bash: ls[!a-m]: command not found...
[mahathi@localhost mahathi148]$ ls [a-v]
ls: cannot access '[a-v]': No such file or directory
[mahathi@localhost mahathi148]$ ls aids
aids
[mahathi@localhost mahathi148]$ ls aids -rwxr-xr--l Lak std 1525 jan10 12:10 aids
ls: invalid line width: 'xr-xr--l'
[mahathi@localhost mahathi148]$ chmod u-wx aids
[mahathi@localhost mahathi148]$ chmod u +rw,g +rw college
chmod: invalid mode: 'u'
Try 'chmod --help' for more information.
[mahathi@localhost mahathi148]$
```

Name : Mahathi A J

RollNo: 241801148

Operating System

1.4 GROUPING COMMANDS

1. The 'semicolon' command:

2. The '&&' operator:

```
[mahathi@localhost mahathi148]$ who;date
mahathi  seat0      2026-03-15 09:50 (login screen)
mahathi  tty2       2026-03-15 09:50 (tty2)
Sunday 15 March 2026 10:36:42 AM IST
[mahathi@localhost mahathi148]$ who&&date
mahathi  seat0      2026-03-15 09:50 (login screen)
mahathi  tty2       2026-03-15 09:50 (tty2)
Sunday 15 March 2026 10:38:15 AM IST
[mahathi@localhost mahathi148]$ who||date
mahathi  seat0      2026-03-15 09:50 (login screen)
mahathi  tty2       2026-03-15 09:50 (tty2)
[mahathi@localhost mahathi148]$ cd rec
bash: cd: rec: Not a directory
[mahathi@localhost mahathi148]$ vi rec
[mahathi@localhost mahathi148]$ head rec
hi, my name is mahathi.
im studying in rajalakshmi engineering college
i'm coming from villivakkam
my age is 19
i'am from aids department
i am a day scholar student and come through bus everyday to college
i am studying OS everyday and also practice.
i am preparing 10 lines to display and create
```

1.5 FILTERS

1. The head filter

2. The tail filter

3. The more filter:

4. The 'grep' command:

5. The 'sort' command:

6. The 'nl' command:

7. The 'cut' command:

Name : Mahathi A J

RollNo: 241801148

Operating System

```
[mahathi@localhost mahathi148]$ tail rec
hi, my name is mahathi.
im studying in rajalakshmi engineering college
i'm coming from villivakkam
my age is 19
i'am from aids department
i am a day scholar student and come through bus everyday to college
i am studying OS everyday and also practice.
i am preparing 10 lines to display and create
[mahathi@localhost mahathi148]$ sort rec
hi, my name is mahathi.
i am a day scholar student and come through bus everyday to college
i'am from aids department
i am preparing 10 lines to display and create
i am studying OS everyday and also practice.
i'm coming from villivakkam
im studying in rajalakshmi engineering college
my age is 19
[mahathi@localhost mahathi148]$ sort -r rec
my age is 19
im studying in rajalakshmi engineering college
i'm coming from villivakkam
i am studying OS everyday and also practice.
i am preparing 10 lines to display and create
i'am from aids department
i am a day scholar student and come through bus everyday to college
hi, my name is mahathi.
[mahathi@localhost mahathi148]$ sort -c rec
sort: rec:3: disorder: i'm coming from villivakkam
[mahathi@localhost mahathi148]$ sort -n rec
hi, my name is mahathi.
i am a day scholar student and come through bus everyday to college
i'am from aids department
i am preparing 10 lines to display and create
i am studying OS everyday and also practice.
i'm coming from villivakkam
im studying in rajalakshmi engineering college
my age is 19
[mahathi@localhost mahathi148]$ sort -m rec
hi, my name is mahathi.
im studying in rajalakshmi engineering college
i'm coming from villivakkam
my age is 19
i'am from aids department
i am a day scholar student and come through bus everyday to college
i am studying OS everyday and also practice.
```

1.5 OTHER ESSENTIAL COMMANDS

1. **free**
2. **top**
3. **ps**
4. **vmstat**
5. **df**
6. **ping**
7. **ifconfig**

Name : Mahathi A J

RollNo: 241801148

Operating System

```
[mahathi@localhost mahathi148]$ vmstat
procs-----memory-----swap-- ----io---- -system-- -----cpu-----
r  b   swpd   free   buff   cache   si   so    bi   bo    in   cs   us   sy   id   wa   st
0  0    3096  221312    2780  739652    0    1   977   122   148  242   1   4   95   0   0
[mahathi@localhost mahathi148]$ df
Filesystem            1K-blocks    Used Available Use% Mounted on
devtmpfs                840224         0    840224   0% /dev
tmpfs                   874308         0    874308   0% /dev/shm
tmpfs                   349724    5564    344160   2% /run
/dev/mapper/cs-root    17756160  6082304   11673856  35% /
/dev/sda1               983040   449000    534040  46% /boot
tmpfs                   174860     104    174756   1% /run/user/1000
[mahathi@localhost mahathi148]$ ping 172.16.4.1
PING 172.16.4.1 (172.16.4.1) 56(84) bytes of data.
^C
--- 172.16.4.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3002ms

[mahathi@localhost mahathi148]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe03:1771  prefixlen 64  scopeid 0x20<link>
    ether 08:00:27:03:17:71  txqueuelen 1000  (Ethernet)
    RX packets 41013  bytes 60983504 (58.1 MiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 4867  bytes 318033 (310.5 KiB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 24  bytes 2514 (2.4 KiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 24  bytes 2514 (2.4 KiB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0
```

```
[root@localhost mahathi148]# traceroute www.rajalakshmi.org
traceroute to www.rajalakshmi.org (192.64.115.115), 30 hops max, 60 byte packets
 1  _gateway (10.0.2.2)  37.904 ms  1.428 ms  1.398 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```